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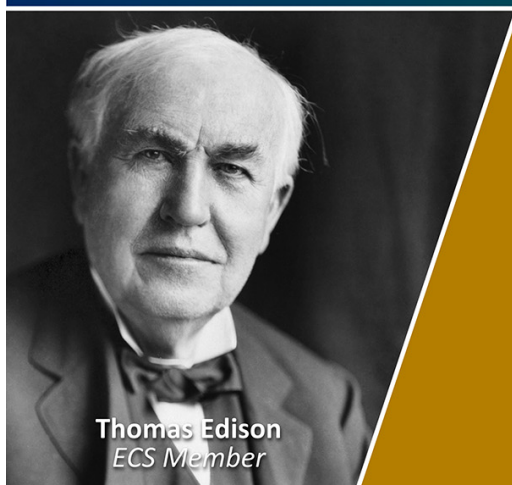
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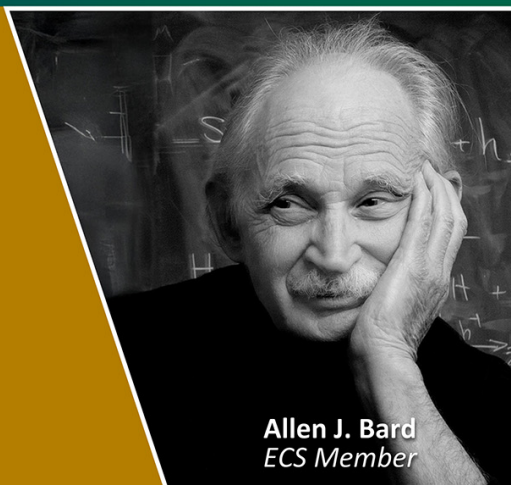


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Implementation of Forest Conservation Policies based on Local Wisdom of the *Ammatoa* Kajang Indigenous Community

S Megawati^{1*} and M A Mahdiannur²

¹Department of Public Administration, Faculty of Social Sciences and Law, Universitas Negeri Surabaya, Surabaya, 60231, Indonesia, ORCID ID: 0000-0002-1798-6879

²Department of Science Education, Faculty of Mathematics and Natural Sciences, Universitas Negeri Surabaya, Surabaya, 60231, Indonesia

^{1*}sucimegawati@unesa.ac.id, ²muhamadmahdiannur@unesa.ac.id

Abstract. The *Ammatoa* Kajang indigenous community maintain cultural authenticity from generation to generation, especially in protecting and preserving customary forests from the threat of damage. This research aims to describe the implementation of forest conservation policies based on local wisdom of the *Ammatoa* Kajang indigenous community. The research method uses a literature/library study approach. Reference documents from 2001-2021. Data analysis methods with data minimization, data visualization, data analysis, and data validation and concluding. The results of the research showed that the Kajang orthodox community in day-to-day forest management is guided by the *Pasang ri Kajang*, which contains rules, implementation and sanctions for all forms of forest utilization and management, monitoring tools and control over all activities related to forestry, positively correlated with forestry policies governance. The concept of customary forest management is to divide the forest into three areas with certain specifications and rules as well as prohibited from destroying flora and fauna. In the end, the forest is sustainable and indigenous community behave sustainably; it can be concluded that the implementation of forest conservation policies based on local wisdom of the *Ammatoa* Kajang indigenous community is going well.

1. Introduction

Indonesia has over 90 million hectares of forest in tropical regions and ranks third-largest area worldwide after Brazil and the Democratic Republic of Congo [1]. In 2020, Indonesia's forest monitoring result revealed that the total of forest area in Indonesia is 95.6 million hectares or 50.9% of Indonesia's total land area, of which 92.5% of the amount of forested area or 88.4 million hectares (218.4 million acres) are in forest areas [2]. The problem of deforestation is a threat, and deforestation is the condition of forest areas that has decreased due to land conventions for infrastructure, settlements, agriculture, mining, and plantations [3]. Information on Indonesia's net logging in 2019-2020 both inside and outside of forest area in Indonesia is 115.5 thousand hectares. The highest logging area occurred in the secondary forest class, which was 104.4 thousand hectares, of which 58.1% or 60.64 thousand hectares were inside the forest zone and the remaining 43.7 thousand hectares or 41.9% were outside the forest zone [2]. The rate of deforestation in Indonesia is still high. According to the Director



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of the Indonesian Forum for the Environment (WALHI), South Sulawesi, the deforestation condition is worrying. In the last decade, the forest area has continued to decrease from 2.1 million hectares to one million hectares [4]. According to Jeffries, the world's conservation strategy is divided into three elements of a conservation strategy, i.e. protection, sustainable use, and sharing the benefits [5]. Various efforts conducted i.e., the preservation of sustainable forest resources and local community-based conservation.

The *Ammatoa* Kajang indigenous community is one of the indigenous community in the Kajang District, Bulukumba Regency, South Sulawesi Province. The *Ammatoa* Kajang indigenous community are still adhere to customary traditions from generation to generation inhabiting the forest. One form of the *Ammatoa* Kajang customary tradition that manifested in the form of protecting and preserving forests. Indonesian Ministry of Environment and Forestry supports the *Ammatoa* Kajang to protect their forest through their tradition that cover an area of $\pm$ 313.99 hectares [6]. The implementation and management of customary forests is conducted by the *Ammatoa* customary community based on *Pasang ri Kajang* and applicable laws and regulations. *Pasang ri Kajang* which is the *Ammatoa* community's way of life which consists of a collection of ancestral mandates has so far been respected and upheld by the Kajang indigenous community [7]. The values hold in the *Pasang* examined sacred by the local, which if they do not implement in their daily activities, they will have a negative effect on the collective life of the ecological and the chaos in the social system.

This research aims to describe the implementation of forest conservation policies based on local wisdom of the *Ammatoa Kajang* indigenous community. Urgency This article is input to policy makers in dealing with deforestation in Indonesia in general and in South Sulawesi Province in particular by using the local wisdom approach of the local community, such as best practice by the *Ammatoa Kajang* indigenous community. The local wisdom approach is contained in written policies or rules upheld and respected by the local community, government, and industry.

2. Method

This research used a literature or library study approach [8]. The data source will be collected from relevant information on forest conservation policies based on local wisdom of the *Ammatoa* Kajang indigenous community through scientific books, scientific journals, research reports, regulations, theses and dissertations, decrees, yearbooks, and encyclopedias both printed and electronic. Reference documents from 2004-2021. The data analysis method for this research is analytical at the time of data collection, capturing the essence of the research aims through the collected sources. Then re-analyze the collected data as raw data that need to be identified with each other. Analytical activities include data minimization, data visualization, data validation, and concluding.

3. Results and discussion

3.1. Implementation of Forest Conservation Policy

The United Nations environmental program mentions the need to integrate different approaches to ensure the broad possibility of conserving biodiversity, recognizes that conservation is strongly influenced by social, cultural, economic, and political factors and encourages cooperation and coordination of policies. and institutional [5]. Local wisdom and the paradigm of sustainable development that are the basis in the preparation of forestry policies will create effective and equitable regulations [9]. The Government of Indonesia issued a policy as stated of the law of the Republic of Indonesia No. 41 of 1999 about forestry which affirms the recognition of customary forests which are no longer state forests. The concept of this rule is the property of customary law communities (also called clan forests, *pertuanan* forest, or as others) are within the scope of land (*ulayat*) rights because they are in one territorial unit of customary law societies, whose demonstrations are based on traditions that live in a community's atmosphere and have a central management agency that is authoritative in the entire territory of its territory.

The Customary Forest Policy opens greater opportunities for village/indigenous communities to access and hold management rights with legal certainty guarantees. It is also supported by the Regional Regulation of the Bulukumba Regency No. 9 of 2015 about the Inauguration, recognition and protection of the rights on the *Ammatoa* Kajang customary law community [10]. The role of the *Ammatoa* Kajang Customary Institution in Preserving Customary Forests i.e., (1) Protect the forest, (2) Supervise the use of forest products, (3) Provide sanctions for those who violate customary rules, (4) Supervise and protect the *paleko'na boronga* (forest blanket) and buffer forest, (5) Determining the time of the activities of traditional rituals and indigenous community in the forest area, and (6) Determining the outer boundaries of the *rambang seppang* and *rambang luara*. The *Ammatoa* as a leader has leadership values, trustworthiness, firmness in upholding the *Pasang* as the principles of community life; and put forward the principle of *abborong* (deliberation) in decision making [11]. The Kajang community are very submissive to their social norms, and one part of their community norm is the concept of forest preservation [12]. The form of the bottom-up approach through the coordination of the local council with customary institutions in protecting the *Ammatoa* Kajang customary forest has now been going well, although the Unity of Action of the Regional Government with customary institutions still need improvement.

3.2. Local Wisdom of the *Ammatoa* Kajang Indigenous Community in Preserving Forests

The belief of Kajang indigenous community considers the Earth as the mother. So, when the wood chopped down, the rain will decline, and the springs will desiccate. When humans take care of nature, nature will also take care of humans [13]. Communities living in the *Ammatoa* Customary Area strongly adhere to the *Pasang ri Kajang* principle, particularly regarding forest conservation, these principles are (a) forests provide a source of life; (b) forests maintain the availability of springs; (c) forests bring rain; and (d) forests cool the environment [7]. In the *Pasang ri Kajang*, all forms of forest use and management are clearly regulated, including being a means of monitoring and controlling all activities related to forestry. The sustainability of the forest in the Kajang cannot be separated from the obedience and respect for the customary laws as outlined in the *Pasang*. In addition to the *Pasang's* teachings, indigenous community also have customary rules called *Patuntung*. The *Patuntung* is a customary law related to efforts to support sustainable forest management. The treatment of the Kajang indigenous community to the forest is not merely to fulfill their daily needs, but also for the sake of maintaining the balance of the ecosystem and its ritual interests [14]. The ability of the *Ammatoa* indigenous community to keep and preserve cultural heritage is due to the togetherness in supporting and practicing the *Pasang*. The knowledge of indigenous community from generation to generation influences bright outlook and efforts in protecting the environment and protecting tropical forests from deforestation [14].

Another effort made by *Ammatoa* to preserve customary forests, by assigning forest guard officers in each direction of the compass consisting of five people (*tau limayya*), namely: *Dalinjok* in the west *ri Balagana*, *Damangassalam* in the south *ri Balangbia*, *Dangempa* in the north *ri Tuli*, *Dakodo* in the east, *ri Teteakak*, and *Tumutung ri Sobbu*, specifically for forests in customary areas. Appointment of guards (*tau limayya*) based on their respective residential areas, as their working area or area by the guidance or instructions from *Pasang ri Kajang* [15]. The customary indigenous forest is divided into three areas, where the similarities between the forests are in the *Ammatoa* customary forest area, and the differences between the three forest areas have different functions. The following is an explanation of each area in the *Ammatoa* Kajang customary forest:

3.2.1. The Sacred Forest (*Borong Karama*)

The first zone of customary forest which according to the tide is prohibited (*kasipalli*) to exploit or disturb flora and fauna in it. The *Borong Karama* can only be accessed by the *Ammatoa* and traditional members if there is a customary ceremony (the *Ammatoa* inauguration ceremony, *Pa'nganroang*). Types of serious violations in sacred forests include: *Ta'bang Kaju* (cutting wood), *Rao Doang* (catching shrimp), *Tatta' Uhe* (cutting down rattan), and *Tunu Bani* (burning bees). There are two types of customary forest (*Borong Karama*) contained in this area, namely *Borong Ilau* and *Borong Iraja*, only

certain community allowed to enter the customary forest area. If the perpetrator is an *Ammatoa* indigenous citizen, they will be tried and given sanctions according to the violation. This person may no longer be allowed to live in the *Ammatoa* customary area. Violations in the sacred forests will be subject to customary sanctions in the form of a whip or a certain amount of money, according to *Ada' Tanayya* (the *Ammatoa* justice system). Perpetrators will be subject to a sanction called *Poko' Ba'bala'*. The *Poko' Ba'bala'* or sanctions for serious violations are the sanctions with the highest fines (IDR 1,200,000) plus a white cloth and wood taken from the sacred forest must be returned [7,12,16].

3.2.2. The Border Forest (*Borong Battasayya*)

The second zone, the border forest is the forest where timber allowed to be harvested if the wood supplies are still available and with permission from the *Ammatoa* (customary leader). The final decision whether the community can take wood in this forest depends on the *Ammatoa*. The wood in this forest is only permitted to build public facilities. In addition, it can be used by community who cannot afford to build houses. Not all types of wood can be logged in the border forest. The size of the wood was decided by the *Ammatoa* himself. The most important requirement is that when you want to log a tree, the person concerned must first plant a tree instead. When one tree is logged, a person plants two trees of the same kind in a location determined by the *Ammatoa*. The logging uses traditional tools in the form of axes or machetes and the logged wood must be removed from the forest by lifting it and not pulling it because it will damage other plants around it. Violations within the border forest area, such as logging without the *Ammatoa*'s permission or cutting more timber than permitted, will be subject to sanctions (*Tangnga Ba'bala'*). This sanction fines the perpetrator which is equivalent to IDR 800,000 plus one roll of white cloth [7,9,17].

3.2.3. The Community Forest (*Borong Luarayya*)

The community forest surrounds the *Amatoan* Community Park which covers an area of ± 100 hectares. From this forest, the community can meet their wood needs with the same needs as at the entrance to the *Borong Battasayya*. The community forest managed by humans, but the usual rules for forest management in the area still apply. The use of this community forest is not allowed. Offenders will be punished with light penalties (*Cappa' Ba'Bala'*) for minor offenses such as damage/collapse of timber in forest areas due to negligence. Violations in the community forest will be punished with a fine equivalent to IDR 400,000 plus a roll of white clothes. This ultimate punishment can also be applied to community who log trees in the *Ammatoa* community garden. Customary sanctions have a significant impact on forest sustainability because they are a form of social punishment that is considered more severe by the *Ammatoa* community than those imposed. Social sanctions are a form of exclusion. This exception also applies to whole families up to the seventh generation [7,12,13,17–19]. The local wisdom of the Kajang customary community can be used as a reference in the formulation of effective and equitable forestry policies [9]. This condition makes the Kajang unique and has its own characteristics to be a lesson for other indigenous community in developing fair and sustainable forestry.

4. Conclusion

Based on the literature study, it can be concluded that the implementation of forest conservation policies based on local wisdom of the *Kajang* indigenous community, which is guided by the *Pasang Kajang*, has succeeded well in preserving the customary forest. Customary forest conservation policies through local wisdom approaches form indigenous community to behave obediently, sustainably, and participate in preserving forests in line with forestry policies in Indonesia. In the future, this can be used as a reference in the formulation of forest policies that are effective, efficient, just, and appropriate in overcoming deforestation.

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